

3. File Handling

What is File Handling?

Used to read/write data into files.

Modes

- `"r"` → Read
- `"w"` → Write
- `"a"` → Append
- `"rb"` → Read binary

Basic Syntax

```
file=open("file.txt","r")
data=file.read()
file.close()
```

Best Practice

```
withopen("file.txt","r")asf:
    data=f.read()
```

Operations

- `read()`
- `readline()`
- `write()`

- append
- file pointer (seek, tell)

PROGRAM

1. Write to file

```
withopen("test.txt","w")asf:  
    f.write("Hello")
```

2. Read file

```
withopen("test.txt","r")asf:  
    print(f.read())
```

3. Append file

```
withopen("test.txt","a")asf:  
    f.write(" World")
```

4. Read line by line

```
withopen("test.txt")asf:  
    forlineinf:  
        print(line)
```

5. Count lines

```
withopen("test.txt")asf:  
    print(len(f.readlines()))
```

6. Count words

```
withopen("test.txt")asf:
```

```
print(len(f.read().split()))
```

7. Copy file

```
withopen("a.txt")asf1,open("b.txt","w")asf2:  
f2.write(f1.read())
```

8. Check file exists

```
importos  
print(os.path.exists("test.txt"))
```

9. Delete file

```
importos  
os.remove("test.txt")
```

10. Write list to file

```
lst= ["a","b"]  
withopen("file.txt","w")asf:  
foriinlst:  
f.write(i+"\n")
```

11. Read specific line

```
withopen("file.txt")asf:  
print(f.readlines()[0])
```

12. File size

```
importos  
print(os.path.getsize("file.txt"))
```

13. Binary write

```
withopen("file.bin", "wb") as f:  
    f.write(b"hello")
```

14. Binary read

```
withopen("file.bin", "rb") as f:  
    print(f.read())
```

15. Use seek

```
withopen("file.txt") as f:  
    f.seek(2)  
    print(f.read())
```

16. Tell position

```
withopen("file.txt") as f:  
    print(f.tell())
```

17. Count characters

```
withopen("file.txt") as f:  
    print(len(f.read()))
```

18. Replace content

```
withopen("file.txt") as f:  
    data=f.read().replace("Hello", "Hi")  
  
withopen("file.txt", "w") as f:  
    f.write(data)
```

19. Write numbers

```
withopen("num.txt", "w") as f:  
    for i in range(5):
```

```
f.write(str(i))
```

20. Merge files

```
withopen("a.txt") as f1, open("b.txt") as f2:  
    withopen("c.txt", "w") as f3:  
        f3.write(f1.read()+f2.read())
```

Checking if File Exists Before Reading

python

```
import os  
  
filename = "hello.txt"  
if os.path.exists(filename):  
    with open(filename, "r") as f:  
        print(f.read())  
else:  
    print("File does not exist.")
```

5.8 Writing Multiple Lines with writelines()

python

```
Pqqqqqqqaaaaaqqqqqqqqqqqqqqqqqqqqqlines = ["Line 1\n", "Line 2\n", "Line 3\n"]  
  
with open("multiline.txt", "w") as f:  
    f.writelines(lines)
```

5.9 Counting Words in a File

python

```
with open("hello.txt", "r") as f:  
    content = f.read()  
    words = content.split()  
    print(f"Word count: {len(words)}")
```

5.10 Copying a File

python

```
with open("hello.txt", "r") as source:  
    with open("copy.txt", "w") as dest:  
        dest.write(source.read())
```

```
print("File copied.")
```

6.1 Writing and Reading JSON Files

python

```
import json

data = {"name": "Alice", "scores": [85, 90, 78]}

with open("data.json", "w") as f:
    json.dump(data, f, indent=2)

with open("data.json", "r") as f:
    loaded = json.load(f)
    print(loaded["name"])
```

6.2 Writing CSV Files

python

```
import csv

rows = [
    ["Name", "Age", "City"],
    ["Alice", 25, "Delhi"],
    ["Bob", 30, "Mumbai"]
]

with open("people.csv", "w", newline="") as f:
    writer = csv.writer(f)
    writer.writerows(rows)
```

6.3 Reading CSV Files

python

```
import csv

with open("people.csv", "r") as f:
    reader = csv.DictReader(f)
    for row in reader:
        print(row["Name"], "-", row["City"])
```

6.4 Exception Handling with Files

python

```
try:
    with open("missing.txt", "r") as f:
        print(f.read())
except FileNotFoundError:
    print("Error: File does not exist.")
```

```
except PermissionError:
    print("Error: No permission to read this file.")
```

6.5 Writing Binary Files

python

```
data = bytes([65, 66, 67, 68]) # ASCII A B C D

with open("binary.bin", "wb") as f:
    f.write(data)

with open("binary.bin", "rb") as f:
    print(f.read()) # b'ABCD'
```

6.6 Using os.path for Safe File Operations

python

```
import os

filepath = "data.json"

if os.path.isfile(filepath):
    size = os.path.getsize(filepath)
    print(f"{filepath} exists. Size: {size} bytes")
else:
    print("File not found.")
```

6.7 Renaming and Deleting Files

python

```
import os

os.rename("old_name.txt", "new_name.txt")
os.remove("new_name.txt")
print("Done.")
```

6.8 Walking a Directory Tree

python

```
import os

for root, dirs, files in os.walk("."):
    for file in files:
        print(os.path.join(root, file))
```

6.9 Using tempfile for Temporary Files

python

```
import tempfile

with tempfile.NamedTemporaryFile(mode="w", delete=False, suffix=".txt") as tmp:
    tmp.write("Temporary data")
    print("Temp file:", tmp.name)
```

6.10 Student Record System (Putting It All Together)

python

```
import json, os

FILE = "students.json"

def load_students():
    if os.path.exists(FILE):
        with open(FILE, "r") as f:
            return json.load(f)
    return []

def save_students(students):
    with open(FILE, "w") as f:
        json.dump(students, f, indent=2)

def add_student(name, grade):
    students = load_students()
    students.append({"name": name, "grade": grade})
    save_students(students)
    print(f"Added {name}.")

def show_students():
    students = load_students()
    for s in students:
        print(f"{s['name']} - Grade: {s['grade']}")

add_student("Alice", "A")
add_student("Bob", "B")
show_students()
```